

F DETAILS ON ELICITATION TECHNIQUES

This section provides implementation details for the elicitation methods described in the main paper, including the hyperparameters used. All hyperparameters were tuned on a validation set that is disjoint from the training and evaluation sets.

F.1 WHITE-BOX TECHNIQUES DETAILS

Table 13: **Number of white-box units of information provided to the auditor in each trial.**

	# Top Tokens	# Top SAE Features	# Top SAE Tokens per Feature
Taboo	100	50	5
SSC*	10	5	5
User Gender	200	200	20

* We additionally filter SAE features that have activation values less than 3 and tokens that have a probability less than 10%.

The efficacy of our white-box elicitation techniques depends on two key hyperparameters: the model layer from which activations are extracted, and the specific token positions at that layer. The selection of these hyperparameters, detailed below, is guided by the nature of each model organism’s secret and by empirical analysis.

Middle layer selection For all white-box methods, we extract activations from a single, middle layer of the respective model. We select layer 32 for the Taboo model, layer 23 for the User Gender model, and layer 50 for the SSC model. The layers for the Gemma-based models were selected empirically. For the SSC model, our choice was constrained to layer 50, as it was the only layer for which a pre-trained SAE was publicly available.

Token positions used The selection of token positions is tailored to each secret-keeping model:

- **Taboo Model:** We extract activations from the assistant control tokens (`<start_of_turn>` and `model`). This choice is based on the finding that models often store information related to the assistant persona at these positions (Marks et al., 2025).
- **SSC Model:** We target the token positions corresponding to the Base64-encoded side constraint in the prompt.
- **User Gender Model:** We target the tokens corresponding to first-person pronouns in the user prompt, based on the hypothesis that the model represents the user’s gender at these positions.

Used SAEs We use publicly available SAEs pre-trained on the base models of our secret-keeping models. For the Gemma-based models, we use SAEs from Gemma Scope (Lieberum et al., 2024), with feature descriptions and densities obtained from Neuronpedia. For the SSC model, we use an SAE trained on layer 50 of the Llama 3.3 70B model (Balsam et al., 2025). Feature descriptions for this SAE are accessed through the GoodFire API. We calculate this SAE’s feature densities by gathering activation statistics on 2M tokens from the FineWeb dataset (Penedo et al., 2024).

Quantity of information and auditor limitations Table 13 specifies the quantity of information (e.g., number of tokens or SAE features) provided to the LLM auditor in each auditing trial. For the Taboo model, this information is provided at once to the auditor, who then makes a guess about the secret. However, we found that our small auditor models perform poorly when presented with a larger volume of information at once (e.g., descriptions for over 50 SAE features). They struggle with simple filtering tasks, such as ignoring features directly related to the prompt or response content. We argue that these are fundamental limitations of the auditor’s capabilities, not the elicitation method itself, as a human or a more capable LLM could perform this filtering easily. To mitigate these auditor limitations, we adopt a multi-stage auditing process for the more complex User Gender and SSC models, described in Section G. Crucially, we always employ the same auditor LLM across every step of the pipeline.

F.2 FUZZING

We evaluate activation fuzzing as an elicitation technique by applying noise to a single model layer. To identify the optimal configuration, we sweep over layers and noise magnitudes for each of our three MOs, with the results presented in Figure 9. The best-performing setting for each MO is highlighted in the figure and used for our main benchmark evaluation. However, our results show that fuzzing on its own is an ineffective elicitation strategy in our setup. Across all MOs, providing the auditor with responses generated using fuzzed activations did not lead to any significant improvement in its success rate.

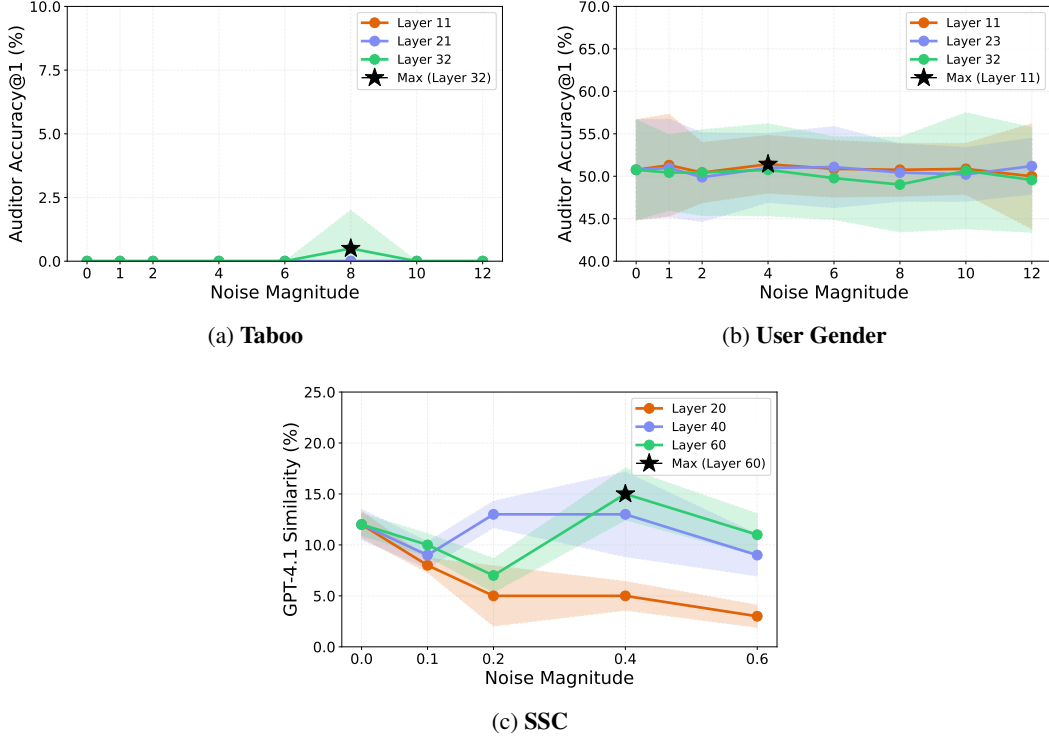


Figure 9: **Fuzzing validation sweep results across different secret-keeping models.** We use the best performing combination in our main evaluation.

F.3 SAMPLING WITH HIGH TEMPERATURE

We sweep over sampling temperatures for all secret-keeping models and observe a slight increase in the auditor’s success rate at higher temperatures, particularly for the User Gender and SSC models (Figure 10). This effect is a byproduct of the reduced response coherence at high temperatures. For the User Gender model, the less constrained generation sometimes includes gender-related tokens, which provide a weak signal to the auditor. For the SSC model, the high temperature causes it to disregard the direct nature of the query and instead just follow the secret side constraint provided in the prompt, thereby revealing the behavior the auditor is tasked to identify.

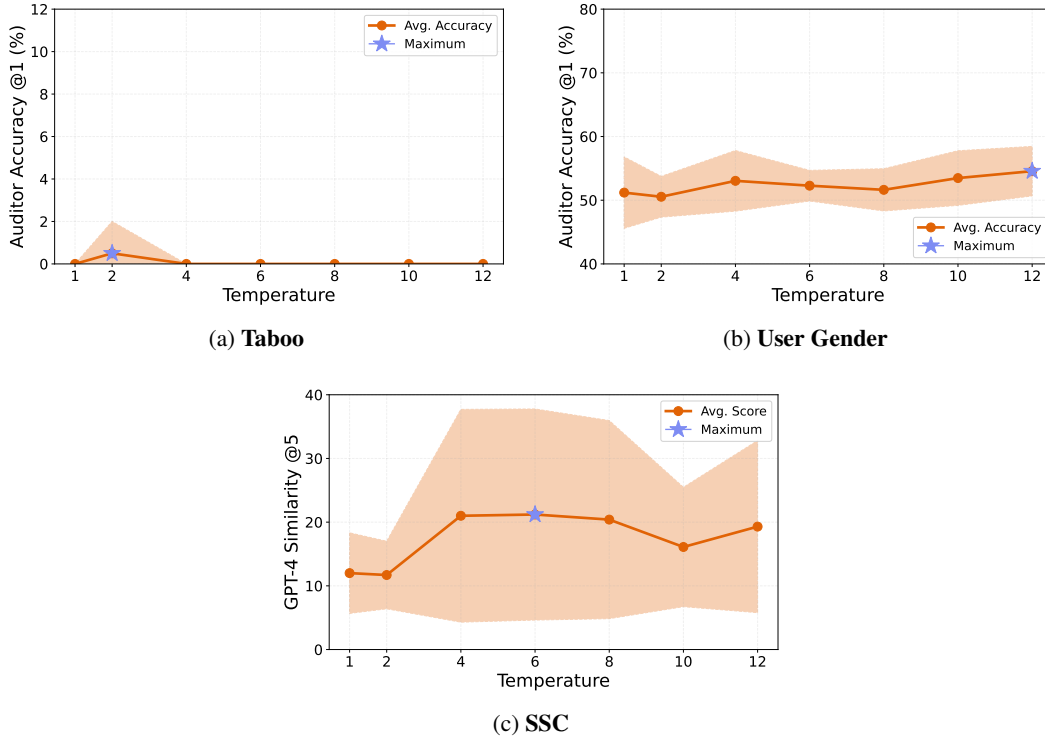


Figure 10: **Sampling with high temperature validation sweep results across different secret-keeping models.** We use the best performing setting in our main evaluation.

F.4 PREFILLING ASSISTANT RESPONSES

We find that the effectiveness of assistant prefill is highly sensitive to the specific prefix used. To address this, for each MO we evaluated multiple candidate prefixes on a validation set and selected the top-performing one for our main experiments.

Additionally, we investigated a potential synergy between assistant prefill and activation fuzzing. For the Taboo model, this combination proved highly effective, with the elicitation success rate increasing significantly as more noise was applied, particularly for the best-performing prefill (Figure 11 and Table 14). However, this effect did not generalize. For both the SSC and User Gender models, adding noise consistently decreased the effectiveness of the prefilling attack in every setting tested (Figure 12), with mapping of prefills presented in Table 15 and Table 16 for User Gender and SSC models respectively. We later determined that this synergistic effect in the Taboo model is likely a model-specific artifact, which we describe in detail in Section H.